



Habib University

CE 321/CS 330 Computer Architecture – Fall 2025

Instructor: Dr. Shehzad Hasan

Time = 50 minutes

Quiz 04

Max Points: 20

Name: _____

ID: _____

Instructions:

- i. **Smart watches, laptops, and similar electronics are strictly NOT allowed.**
- ii. **Answer sheets should contain all steps, working, explanations, and assumptions.**
- iii. Attempt the quiz on clean papers with black/blue ink.
- iv. Print your name and HU ID on all sheets.
- v. Turn in your question paper along with your answer sheets.
- vi. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	<i>Explain</i> the role of ISA in modern processors and instruction encodings and assembly language programming	
CLO 2	<i>Explain</i> the architecture and working of a single cycle processor	
CLO 3	<i>Design</i> the architecture to mitigate issues of a pipelined processor	
CLO 4	<i>Analyze the</i> performance of cache operations	✓

Question 1 [8 points]:

[4 points] Consider a two-way set associative cache with a total of 4 blocks (lines) each of size 16 bytes (4-words). Assume an LRU (Least Recently Used) replacement policy. For the memory references given below, fill in the following table. Also specify the LRU block.

Memory References (word numbers): 0, 2, 8, 7, 17, 13, 6, 20, 1, 3, 11, 12

Word Number	MM Block Number	Set Assigned	Hit / Miss	Set 0		Set 1	
0	0	0	Miss	[0,1,2,3]			
2	0	0	Hit	[0,1,2,3]			
8	2	0	Miss	[0,1,2,3]*	[8,9,10,11]		
7	1	1	Miss	[0,1,2,3]*	[8,9,10,11]	[4,5,6,7]	
17	4	0	Miss	[16,17,18,19]	[8,9,10,11]*	[4,5,6,7]	
13	3	1	Miss	[16,17,18,19]	[8,9,10,11]*	[4,5,6,7]*	[12,13,14,15]
6	1	1	Hit	[16,17,18,19]	[8,9,10,11]*	[4,5,6,7]	[12,13,14,15]*
20	5	1	Miss	[16,17,18,19]	[8,9,10,11]*	[4,5,6,7]*	[20,21,22,23]
1	0	0	Miss	[16,17,18,19]*	[0,1,2,3]	[4,5,6,7]*	[20,21,22,23]
3	0	0	Hit	[16,17,18,19]*	[0,1,2,3]	[4,5,6,7]*	[20,21,22,23]
11	2	0	Miss	[8,9,10,11]	[0,1,2,3]*	[4,5,6,7]*	[20,21,22,23]
12	3	1	Miss	[8,9,10,11]	[0,1,2,3]*	[12,13,14,15]	[20,21,22,23]*

Miss rate : $9/12 = 0.75$

- a) **[4 points]** Assume now that the same cache is implemented as a fully-associative cache i.e. 4 lines each of size 16 bytes but with a FIFO replacement policy. Consider the same word number sequence, fill in the following table, show the FIFO block and calculate the miss rate?

Memory References (word numbers): 0, 2, 8, 7, 17, 13, 6, 20, 1, 3, 11, 12

Word Number	Hit/Miss				
0	Miss	[0,1,2,3]*			
2	Hit	[0,1,2,3]*			
8	Miss	[0,1,2,3]*	[8,9,10,11]		
7	Miss	[0,1,2,3]*	[8,9,10,11]	[4,5,6,7]	
17	Miss	[0,1,2,3]*	[8,9,10,11]	[4,5,6,7]	[16,17,18,19]
13	Miss	[12,13,14,15]	[8,9,10,11]*	[4,5,6,7]	[16,17,18,19]
6	Hit	[12,13,14,15]	[8,9,10,11]*	[4,5,6,7]	[16,17,18,19]
20	Miss	[12,13,14,15]	[20,21,22,23]	[4,5,6,7]*	[16,17,18,19]
1	Miss	[12,13,14,15]	[20,21,22,23]	[0,1,2,3]	[16,17,18,19]*
3	Hit	[12,13,14,15]	[20,21,22,23]	[0,1,2,3]	[16,17,18,19]*
11	Miss	[12,13,14,15]*	[20,21,22,23]	[0,1,2,3]	[8,9,10,11]
12	Hit	[12,13,14,15]*	[20,21,22,23]	[0,1,2,3]	[8,9,10,11]

Miss rate : $8/12 = 0.67$

Question 2 [8 points]:

A four-way set-associative cache with 256 sets has been proposed for an embedded RISC-V processor. Each block holds 16 words. The memory address space is 1 GB (30-bit address), and the memory is byte-addressable. (Note: Each word contains 4 bytes)

- a) [4 points] How many bits will be allocated for tag, index(set), and block offset fields?

Tag	Index (Set)	Block Offset
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(Note: Block offset also includes byte offset in its least significant bits)

The block address specifies the number of bytes in a block.

Block Size = 16 words = 64 bytes so 6 bits will be required for block offset.

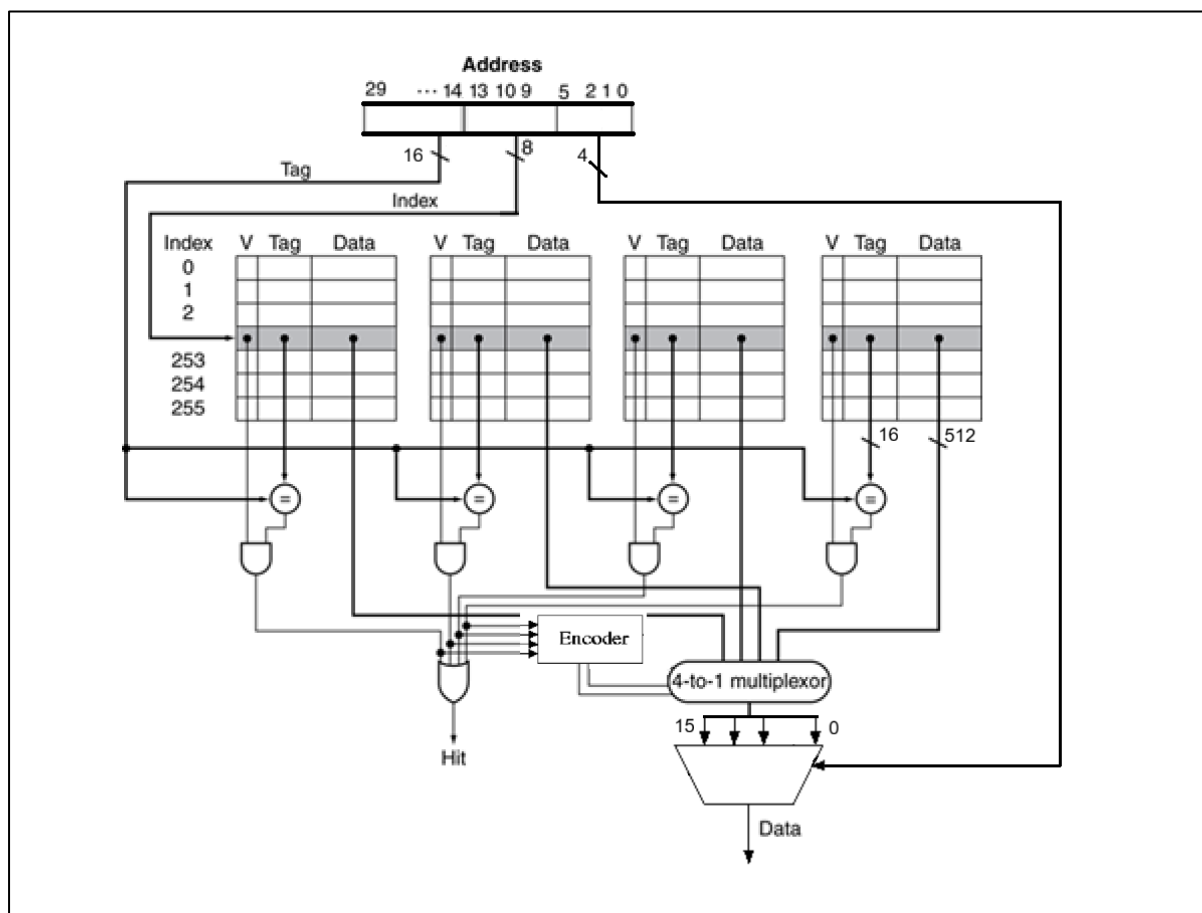
Index field specifies the number of sets

Set = 256 so 8 bits are required for index field

As main memory address is 30 bits then remaining bits are for Tag

Tag bits = $30 - 8 - 6 = 16$ bits

- b) [4 points] Draw a diagram for the cache organization specifying the relevant multiplexer(s) and other logic.



Question 3 [4 points]:

- a) [2 points] Assume the miss rate of an instruction cache is 6% and the miss rate of the data cache is 10%. If a processor has a CPI of 1.35 without any memory stalls, and the miss penalty is 70 cycles for all misses, determine how much faster a processor would run with a perfect cache that never missed.

Assume the frequency of all loads and stores is 42%.

- For a perfect cache

There are no memory stall clock cycles

$$\begin{aligned}\text{CPU time with perfect cache} &= \text{IC} \times \text{CPI} \times \text{clock cycle time} \\ &= I \times 1.35 \times T\end{aligned}$$

- For memory with stalls

Memory-stall clock cycles = Instruction cache stall cycles + Data cache stall cycles

Instruction cache stall cycles = Instruction accesses/program $\times$ I-cache miss rate $\times$ miss penalty

$$= I \times 0.06 \times 70$$

$$= I \times 4.2$$

Data cache stall cycles = Data accesses/program $\times$ D-cache miss rate $\times$ miss penalty

$$= I \times 0.42 \times 0.1 \times 70$$

$$= I \times 2.94$$

Memory-stall clock cycles = Instruction cache stall cycles + Data cache stall cycles

$$= I \times 4.2 + I \times 2.94 = I \times 7.14$$

CPU time with stalls = (IC $\times$ CPI + memory stalls) $\times$ clock cycle time

$$= (I \times 1.35 + I \times 7.14) \times T = I \times 8.49 \times T$$

- Speed up

$$\frac{\text{CPU time with stalls}}{\text{CPU time with perfect cache}} = \frac{I \times 8.49 \times T}{I \times 1.35 \times T} = \frac{8.49}{1.35} = 6.289$$

- b) [2 points] Suppose we include an L2 cache in the design to improve the miss penalty. The L2 cache however has a local miss rate of 20% and a hit time of 10 cycles? What would be the effect on CPU cycles with this new cache?

Instruction cache stall cycles = Instruction accesses/program $\times$ I-cache miss rate $\times$

(L2 hit cycles + L2 miss rate $\times$ miss penalty in cycles)

$$= I \times 0.06 \times (10 + 0.2 \times 70)$$

$$= I \times 1.44$$

Data cache stall cycles = Data accesses/program $\times$ D-cache miss rate $\times$

(L2 hit cycles + L2 miss rate $\times$ miss penalty in cycles)

$$= I \times 0.42 \times 0.1 \times (10 + 0.2 \times 70)$$

$$= I \times 1.008$$

$$\text{Old CPU cycles} = (I \times 1.35 + I \times 7.14)$$

$$\text{New CPU cycles} = (I \times 1.35 + I \times (1.44 + 1.008))$$

$$\text{Speedup} = 8.49/3.798 = 2.235 \text{ times faster}$$